Gabor Granular Synthesis & Time-Stretching Math

Mathematical Models of the `[sampler~]` Granular Engine

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Abstract

This paper formulates the Gabor granular time-stretching algorithms, grain window envelopes, stochastic Poisson grain distributions, and overlap-add (OLA) sum equations powering the `[sampler~]` audio node.

1 Gabor Grain Representation

A single audio grain $g_k(t)$ of length L_k centered at buffer offset τ_k with pitch playback speed S_k is defined by modulating an audio sample buffer $x(t)$ with a smooth window envelope $w(t)$:

$$g_k(t) = w\left(\frac{t - t_{k,\text{start}}}{L_k}\right) \cdot x\left(\tau_k + S_k \cdot (t - t_{k,\text{start}})\right) \tag{1}$$

where $t_{k,\text{start}}$ is the physical time instant when grain k is spawned.

2 Grain Windowing Envelope Functions

To eliminate boundary click artifacts at grain edges ($t = 0$ and $t = L_k$), window function $w(\eta)$ for $\eta \in [0, 1]$ is selected:

2.1 Hanning Window Envelope

$$w_{\text{Hanning}}(\eta) = \frac{1}{2}\left(1 - \cos\left(2\pi\eta\right)\right) = \sin^2(\pi\eta) \tag{2}$$

2.2 Gaussian Window Envelope

$$w_{\text{Gaussian}}(\eta) = \exp\left(-\frac{(\eta - 0.5)^2}{2\sigma^2}\right), \quad \text{where } \sigma \approx 0.16 \tag{3}$$

3 Stochastic Poisson Grain Spawning Distribution

The inter-grain arrival interval Δt_{spawn} is governed by a Poisson point process with mean grain density rate λ (grains per second):

$$P(\Delta t_{\text{spawn}} = \delta) = \lambda e^{-\lambda\delta} \tag{4}$$

In discrete time, next grain spawn time n_{next} is randomized using uniform random variate $U \in (0, 1)$:

$$n_{\text{next}} = n_{\text{current}} + \left\lfloor -\frac{f_s}{\lambda}\ln(U) \right\rfloor \tag{5}$$

4 Overlap-Add (OLA) Grain Summation

The continuous output signal $y(t)$ is generated by summing all active overlapping grains $g_k(t) \in \mathcal{G}_{\text{active}}$:

$$y(t) = \sum_{k \in \mathcal{G}_{\text{active}}} g_k(t) \tag{6}$$

Under uniform grain overlap with Hanning windowing, constant power energy normalization $\sum w_k(t)^2 = 1.0$ is maintained, preserving original dynamic loudness across variable grain densities.